

Hall Ticket Number:

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Code No. : 13258 N/O

VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS), HYDERABAD

Accredited by NAAC with A++ Grade

B.E. III-Semester Main & Backlog Examinations, February/March-2023**Computer Architecture**

(Common to CSE & AIML)

Time: 3 hours

Max. Marks: 60

Note: Answer all questions from **Part-A** and any **FIVE** from **Part-B****Part-A (10 × 2 = 20 Marks)**

Q. No.	Stem of the question	M	L	CO	PO
1.	Compare and contrast computer architecture with computer organization?	2	2	1	1
2.	Draw and explain the block diagram and timing diagram for the statement. P1: R2 ← R1	2	1	1	2
3.	List out the advantages and disadvantages of hard-wired and micro-programmed control unit?	2	1	2	1
4.	Explain the following RTL concerned to instruction BSA? D ₅ T ₄ : M[AR] ← PC, AR ← AR+1 D ₅ T ₅ : PC ← AR, SC ← 0	2	2	2	2
5.	What is parallel processing? Explain Flynn's classification of computer?	2	1	3	1
6.	A non-pipeline system takes 100ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 20ns. Determine the speedup ratio of the pipeline for 200 tasks?	2	2	3	2
7.	What are the values of input ready, output ready, and control bits F1 through F4, when (a) The buffer is empty? (b) The buffer is full?	2	1	4	1
8.	Define bit-oriented protocol? Determine the minimum and maximum number of bits that a frame must have in the bit-oriented protocol?	2	2	4	2
9.	Determine an average access time of a computer with cache access time of 100 ns, a main memory access time of 1000 ns, and a hit ratio of 0.9?	2	2	5	2
10.	State Amdahl's Law? If a system has a single bottleneck that occupies 25% of the total execution time, and we add 5 more processors to the system, find the speedup?	2	2	5	2
Part-B (5 × 8 = 40 Marks)					
11. a)	Design 4-bits adder/subtractor using full adder with suitable gates?	4	3	1	3
b)	Let the contents of 8-bit registers A, B, C, and D initially are 11110010, 11111111, 10111001, and 11101010 respectively. Determine the 8-bit values in each register after performing following sequence of microoperations: A ← A + B // Add B to A C ← C ^ D // AND D to C B ← B + 1 // Increment B A ← A - C // Subtract C from A	4	2	1	2
12. a)	Explain control memory and the associated hardware needed for selecting the next microinstruction address with neat block diagram?	4	1	2	1
b)	A Computer uses a memory unit with 256K word of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, and a register code part to specify one of 64 registers and an address part. How many bits are there in operation code, the register code part, and the address part?	4	3	2	3

13. a)	What are the addressing modes? Explain different types of addressing modes with suitable examples?	4	1	3	1
b)	Draw the block diagram of an 8-bit ALU with a 4-bit status register? Explain four status bits (C, S, Z, and V) in brief? Determine the status of 4-bits register after execution of following instructions. Assume that the value of 8-bit register R1 in each case is 9C (Hexadecimal)? (a) Add immediate operand 1B to R1 (b) Subtract immediate operand 7A from R1	4	3	3	3
14. a)	Explain Asynchronous Communication Interface or Universal Asynchronous Receiver-transmitter (UART) with neat block diagram and function table?	4	2	4	2
b)	Obtain the truth table of an 8 x 3 priority encoder. Assume that the three outputs xyz from the priority encoder are used to provide a vector address of the form 101xyz00. Use the eight vector addresses starting from the one with the highest priority?	4	3	4	3
15. a)	Explain Segmented-Page Mapping with neat block diagram? Explain the format of a typical segment descriptor?	4	2	5	3
b)	Consider a machine with a clock frequency of 3 GHz executing 4000 instructions with an average CPI of 3. Suppose we use a new compiler on the same program for which the new instruction count is 7000, and a new CPI is 1.8, running on a faster machine with a clock rate of 4 GHz. What will be the speedup achieved?	4	3	5	3
16. a)	Explain typical one stage Arithmetic Logic Shift Unit (ALU) with block diagram and function table?	4	2	1	2
b)	What is interrupt? Explain interrupt cycle with neat flowchart?	4	2	2	1,2
17.	Answer any <i>two</i> of the following:	4	3	3	3
a)	Perform the multiplication of (+9) and (-13) using Booth's multiplication algorithm?	4	3	4	3
b)	A DMA controller transfers 16 bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 2400 characters per second. The CPU is fetching and executing instructions at an average rate of 1 million instructions per second. By how much will the CPU be slowed down because of the DMA transfer?	4	3	5	3
c)	A computer employs RAM chips of 256 x 8 and ROM chips of 1024 x 8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units, each with four registers. A memory-mapped I/O configuration is used. The two highest order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. a. How many RAM and ROM chips are needed? b. Draw a memory-address map for the system. c. Give the address range in hexadecimal for RAM, ROM, and interface	4	3	5	3

M : Marks; L: Bloom's Taxonomy Level; CO; Course Outcome; PO: Programme Outcome

i)	Blooms Taxonomy Level – 1	20%
ii)	Blooms Taxonomy Level – 2	40%
iii)	Blooms Taxonomy Level – 3 & 4	40%
